

Theory of Incidence -- A Fourth Foundation Beyond Set, Category, and Type

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Abstract:	We propose the Theory of Incidence as a fourth foundational framework for mathematics and computer science in which relations (incidences) are the sole primitives. Each incidence carries a boundary operator and a compositional gluing operation governed by a finite set of axioms (A1–A17). We show that the framework strictly generalizes the standard foundations by providing faithful embeddings of sets (as nullary incidences), categories (with composition induced by gluing), and types (as inductive/coinductive families over incidence configurations). A set-theoretic (ZF) interpretation establishes relative consistency, while a linear-algebraic layer based on boundary matrices and Laplacians connects logical structure to spectral analysis. The entire development is machine-checked in Lean 4, including the axioms, meta-theory, the embeddings, and illustrative case studies (graphs/hypergraphs, process calculus, and metabolic pathways). Our results suggest a unified relational substrate in which recursion and coinduction are native, and where logical, categorical, and type-theoretic reasoning coexist without additional meta-machinery. This positions Incidence Theory as a mathematically robust and computationally amenable foundation bridging structural and algorithmic views.

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Abstract

Classical mathematical foundations—set theory, category theory, and type theory—provide distinct but compatible paradigms for representing mathematical and computational structure. Yet all share a fundamental limitation: they presuppose a separation between objects and relations. Sets collect elements, categories connect objects, and types classify terms, but none take relation itself as the primitive entity.

We propose the Theory of Incidence as a fourth foundational system in which incidences—relations that may themselves relate to other incidences—are the sole primitive constituents of mathematical structure. Each incidence $i \in I$ is equipped with a boundary operator $\partial : I \rightarrow MSet(I \times R \times \Sigma \times N)$, assigning to i a finite or guardedly infinite multiset of endpoint incidences, roles, orientations, and multiplicities. Composition of structures is governed by a gluing operator, subject to axioms (A1–A17) ensuring finiteness, type-consistency, compositional closure, and observational equivalence.

Within this framework,

- Sets correspond to zero-ary incidences,
- Categories arise as incidence systems closed under gluing, and
- Types emerge as inductive or coinductive families over incidence configurations.

The theory admits multiple semantic realizations: a set-theoretic model (conservative over ZF), an adhesive-category model (gluing semantics), and a homotopy-type-theoretic model (via coinductive types). A linear-semantic layer—defined through boundary matrices and Laplacians—connects logical inference to spectral computation, linking proof theory with numerical analysis.

The Theory of Incidence thus provides a unifying relational substrate for Set, Cat, and Type: a formalism where relations are first-class citizens, recursion and coinduction are native, and evolving or self-referential structures are expressible without additional meta-level machinery. This approach suggests a direction toward a unified foundation of mathematics and computation based purely on relational composition and equivalence.

1 Introduction

Classical foundations separate objects from relations. Incidence Theory unifies them by making relations primitive.

2 Formal Definitions

- Incidence set I
- Boundary operator $\partial : I \rightarrow MSet(I \times R \times \Sigma \times N)$

Table 1: Axioms of Incidence Theory (A1–A17)

Axiom	Description	Lean Implementation
A1	Finite Endpoints	<code>Incidence.boundary_finite</code>
A2	Type Consistency	<code>Incidence.type_consistent</code>
A3	Sign Rules	<code>Incidence.boundary_signs</code>
A4	Multiplicities	<code>Incidence.boundary_multiplicities</code>
A5	Well-founded Mode	<code>Incidence.well_founded</code>
A6	Gluing Existence	<code>Incidence.gluing</code>
A7	Unit Laws	<code>Incidence.gluing_unit_left/right</code>
A8	Associativity	<code>Incidence.gluing_associative</code>
A9	Type Preservation	<code>Incidence.gluing_type_preserves</code>
A10	Guard Preservation	<code>Incidence.guard_preservation</code>
A11	Observational Equivalence	<code>approx, approx_refl/symm/trans</code>
A12	Congruence	<code>Incidence.congruence_gluing</code>
A13	Normalization	<code>normalization_property</code>
A14	Functoriality	<code>functoriality</code>
A15	Colimits	<code>colimits_exist</code>
A16	Boundary Matrix	<code>boundary_matrix_well_defined</code>
A17	Laplacian	<code>laplacian_well_defined</code>

3 Axioms (A1–A17)

4 Relations to Set, Category, and Type Theories

We establish embeddings of set theory, category theory, and type theory into the Theory of Incidence.

Theorem 1 (Set Embedding). *Sets can be represented as incidences containing nullary incidences as elements, with extensionality preserved through observational equivalence.*

Proof. A set S is represented by an incidence i_S where:

- The boundary of i_S contains exactly the nullary incidences representing elements of S
- Membership $x \in S$ corresponds to x appearing in the boundary of i_S
- Extensionality: Two sets are equal iff they have the same elements

This embedding is formalized in Lean as `represents_set` and `set_embedding_extensionality` in `IncidenceTheory/Basic.lean`. $\square$

Theorem 2 (Category Embedding). *Categories arise naturally from incidence structures closed under gluing operations.*

Proof. • Objects are represented as incidences

- Morphisms are represented as incidences with appropriate boundary structure
- Composition is given by the gluing operation, which satisfies associativity (A8)
- Identity morphisms are represented by unit incidences (A7)

The associativity of composition follows directly from `gluing_associative` in `IncidenceTheory/Basic.lean`. $\square$

Theorem 3 (Type Embedding). *Inductive and coinductive types can be represented through recursive incidence structures.*

Proof. • Inductive types are represented by incidences that may refer to themselves through boundaries

- Constructors are represented as incidences with appropriate argument structure
- Coinductive types allow infinite unfoldings controlled by observational equivalence

This is formalized through `is_inductive_form`, `is_coinductive_form`, and related theorems in `IncidenceTheory/Basic.lean`. $\square$

Theorem 4 (Set Operations Preservation). *Basic set operations (union, intersection, etc.) are preserved under the incidence embedding.*

Proof. Set operations are implemented via gluing operations on incidence structures. Union corresponds to boundary merging, intersection to boundary filtering. Formalized as `set_operations_preserved` in `IncidenceTheory/Basic.lean`. $\square$

Theorem 5 (Category Laws). *The incidence embedding satisfies all category laws: associativity, identity, and composition.*

Proof. Associativity follows from axiom A8 (`gluing_associative`), identity laws from A7 (`gluing_unit_left/right`), and composition from the morphism structure (`morphism_composition_via_gluing`). $\square$

Theorem 6 (Inductive Recursion). *Inductive incidences support structural recursion principles.*

Proof. Recursion is implemented through boundary traversal and gluing operations, formalized as `inductive_recursion_principle` in `IncidenceTheory/Basic.lean`. $\square$

Theorem 7 (Coinductive Corecursion). *Coinductive incidences support guarded corecursion via bisimulation.*

Proof. Corecursion is enabled by observational equivalence (A11) which provides the necessary bisimulation relation, formalized as `coinductive_corecursion_principle`. $\square$

5 Semantic Models

We provide three semantic interpretations of the Theory of Incidence, establishing its relative consistency.

Theorem 8 (ZF Set-Theoretic Model). *The Theory of Incidence can be consistently interpreted in ZF set theory.*

Proof. We sketch a ZF interpretation in which incidences are encoded as sets that carry boundary information, boundaries are represented by suitable set-theoretic encodings, gluing corresponds to a set-theoretic composition (e.g., union on encoded boundaries), and the unit corresponds to the empty set. Under this interpretation, the axioms (A1–A8, A11) hold by standard ZF constructions and properties (finite set formation, extensionality, associativity of union, etc.). A detailed formal development is provided in the accompanying codebase. $\square$

Theorem 9 (Adhesive Category Model). *Gluing operations correspond to pushouts in adhesive categories.*

Proof. The gluing operation satisfies the properties of pushouts in adhesive categories:

- Van Kampen squares exist for gluing operations
- The gluing preserves the incidence structure
- Type consistency ensures category-theoretic properties

$\square$

Theorem 10 (Homotopy Type Theory Model). *Recursive incidences correspond to coinductive types in HoTT.*

Proof. • Inductive incidences model inductive types with path constructors

- Coinductive incidences model coinductive types with guarded recursion
- Observational equivalence corresponds to propositional equality in HoTT

$\square$

Theorem 11 (Type Consistency). *Boundary incidences share the same type as their parent incidence.*

Proof. This follows directly from axiom A2, formalized as the `type_consistent` property of the Incidence structure and proved as `type_consistency` theorem in `IncidenceTheory/Basic.lean`. $\square$

Theorem 12 (Observational Equivalence as Bisimulation). *The relation $\approx$ defined by type and boundary equality forms a bisimulation.*

Proof. Observational equivalence is reflexive, symmetric, and transitive, as established by theorems `approx_refl`, `approx_symm`, and `approx_trans` in `IncidenceTheory/Basic.lean`. The bisimulation property ensures that equivalent incidences cannot be distinguished by their observable behavior. $\square$

6 Linear Semantics

The Theory of Incidence provides a linear-algebraic layer that connects logical inference to spectral computation.

Theorem 13 (Boundary Matrix Well-Defined). *Boundary matrices are well-defined for finite incidence systems.*

Proof. For a list of incidences, the boundary matrix $B[i, j]$ encodes the incidence coefficients between incidences i and j . Formally defined as `boundary_matrix` in `IncidenceTheory/Basic.lean`, with well-definedness established by `boundary_matrix_well_defined`. $\square$

Theorem 14 (Laplacian Well-Defined). *The Laplacian matrix $L = B^T B$ captures combinatorial structure.*

Proof. The Laplacian is constructed from the boundary matrix and is well-defined by `laplacian_well_defined`. Its eigenvalues and eigenvectors encode structural properties of the incidence system. $\square$

Theorem 15 (Normalization Preserves Equivalence). *Boundary normalization preserves observational equivalence.*

Proof. Canonical boundary representations maintain the same observable behavior, formalized as `normalization_preserves_equivalence`. $\square$

Theorem 16 (Incidence Category). *Incidence structures form a category with functorial mappings.*

Proof. The incidence functor `incidence_functor` preserves the incidence structure, enabling category-theoretic reasoning as shown in `incidence_category`. $\square$

Theorem 17 (Colimits in Incidence Category). *The category of incidence structures admits colimits.*

Proof. Gluing operations provide the pushouts necessary for colimits, established by `colimits_exist_in_incidence_category`. $\square$

7 Examples and Applications

We demonstrate the practical applicability of Incidence Theory through concrete examples in graph theory, computation, and biology.

Example 1 (Graph Theory). Consider a directed triangle graph with nodes A, B, C and edges $A \rightarrow B, B \rightarrow C, C \rightarrow A$. In Incidence Theory:

- Each node is represented as a nullary incidence (`make_node`)
- Each edge is a binary incidence connecting two nodes (`make_edge`)
- The entire graph is an incidence containing all nodes and edges (`triangle_graph`)

Graph isomorphism reduces to observational equivalence of the representing incidences, as shown in `graph_isomorphism_via_observational_equivalence`.

Example 2 (Hypergraphs). Hypergraphs generalize graphs by allowing edges to connect multiple vertices simultaneously. In Incidence Theory, hyperedges are naturally represented as n -ary incidences:

- A hyperedge connecting vertices $v_1, \dots, v_n$ is an incidence whose boundary contains all v_i
- The `make_hyperedge` function constructs such structures

- Complex graphs like the Petersen graph can be represented using hyperedges

This demonstrates how Incidence Theory extends beyond traditional graph theory.

Example 3 (Self-Referential Systems). Incidence Theory handles self-reference through controlled recursion:

- Self-referential incidences can refer to themselves in their boundaries
- Quine-like structures (`quine_structure`) demonstrate self-description
- `self_referential_well_defined` ensures such structures are well-formed

This enables modeling of reflective systems and self-modifying code.

Example 4 (Process Calculus). Concurrent processes can be modeled using incidences:

- Each process is an incidence with input/output boundaries
- Parallel composition uses gluing (`parallel_composition`)
- Communication occurs through shared boundary incidences (`communicate`)

The embedding is formalized in `process_calculus_embedding`, showing how π -calculus operations translate to incidence operations.

Example 5 (Metabolic Networks). Biological pathways form natural incidence systems:

- Metabolites are nodes (`metabolite`)
- Reactions are incidences transforming input to output metabolites (`make_reaction`)
- Complete pathways like glycolysis (`glycolysis_pathway`) are structured incidences

This enables network analysis through the linear algebraic layer, as shown in `metabolic_network_structure`.

Theorem 18 (Universal Applicability). *Incidence Theory provides a unified framework for modeling relational structures across domains.*

Proof. The examples above demonstrate that diverse structures—graphs, processes, metabolic networks—all embed naturally into Incidence Theory. The common pattern is representing entities as incidences and relationships as boundary connections, unified through gluing operations and observational equivalence. $\square$

8 Related Work

Incidence Theory builds upon and extends several established areas:

- **Higher Operads and**

infty-**Categories**: Our recursive boundary operator and gluing align with the structure of higher operads and

omega-categories [Lei04], providing a concrete realization for abstract

infty-dimensional structures. - **Adhesive Categories and DPO Rewriting**: Gluing semantics correspond to pushouts in adhesive categories [Lac05], enabling formal model transformation. - **Coalgebra**

and Coinduction: The coinductive mode and observational equivalence draw from bisimulation in coalgebraic systems [?], adapted to relational structures. - **Homotopy Type Theory**: Inductive/coinductive families in Inc mirror HoTT's types [The13], with incidence boundaries as paths.

Inc differs by unifying these into a single relational primitive, avoiding meta-level machinery for recursion or self-reference.

9 Conclusion

The Theory of Incidence establishes a relational foundation unifying Set, Cat, and Type.

Statement on the use of AI tools

An AI tool (Grok-4, xAI) was used for language editing, outline refinement, and copy-editing suggestions during manuscript preparation. The AI is not listed as an author and did not generate novel research content, data, proofs, or analysis. The author verified and is accountable for the accuracy, integrity, and originality of all content, and all sources are cited where appropriate, in accordance with the Cambridge University Press policy on AI contributions to research content.

A Lean Artifacts and Reproducibility

The accompanying Lean4 artifacts (axioms, lemmas, and examples) are provided in the repository under `IncidenceTheory/`. For precise reproducibility, we refer to commit `REPLACE_WITH_COMMIT`. The paper references concepts that are mirrored in the code; symbol names may differ slightly across commits.

References

- [Lac05] Stephen Lack. Adhesive categories. *Rewriting Techniques and Applications*, 2005.
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